

Data Analyst Securitisation

1. Introduction :

Banking and Securitisation

The banking sector provides financial products such as personal loans, home loans, auto loans, and business loans to customers. As loan portfolios grow, banks need mechanisms to manage credit risk and maintain liquidity. One of the most effective methods is **securitisation**, where a pool of loans is packaged into financial securities and sold to investors.

Securitisation transfers part of the credit risk from the originating bank to investors while providing immediate liquidity. Investors receive returns based on repayments generated by the underlying loan portfolio. Therefore, continuous monitoring of portfolio quality, delinquency, expected losses, and recovery performance is essential.

This project develops a Power BI–based **Securitisation Risk Analytics Platform** to monitor portfolio performance, calculate expected credit losses, perform stress testing, and provide investor reporting through interactive dashboards.

IFRS 9

IFRS 9 (International Financial Reporting Standard 9) is an accounting standard governing the classification, measurement, and impairment of financial instruments.

A key feature of IFRS 9 is the **Expected Credit Loss (ECL)** model, which requires financial institutions to estimate future credit losses before actual defaults occur. Loans are classified into three impairment stages:

- **Stage 1:** Performing assets with low credit risk.
- **Stage 2:** Assets with a significant increase in credit risk.
- **Stage 3:** Credit-impaired or defaulted assets.

This project incorporates IFRS 9 concepts by calculating Expected Credit Loss and analyzing loan portfolios according to credit risk.

Expected Credit Loss (ECL)

Expected Credit Loss (ECL) represents the estimated financial loss a lender expects to incur due to borrower defaults.

The standard formula is:

$$\text{ECL} = \text{PD} \times \text{LGD} \times \text{EAD}$$

Where:

- **PD:** Probability of Default
- **LGD:** Loss Given Default
- **EAD:** Exposure at Default

Power BI DAX measures are used to calculate ECL across different loan types, regions, and portfolio segments.

Delinquency

Delinquency refers to overdue loan payments. It is measured using **Days Past Due (DPD)**.

Typical DPD buckets include:

- Current (0 Days)
- 1–30 Days
- 31–60 Days
- 61–90 Days
- 90+ Days

Monitoring delinquency helps identify deteriorating loan quality and enables proactive risk management.

Static Pool Analysis

Static Pool Analysis evaluates loans grouped by their origination period (vintage). Each loan pool is monitored over time to measure cumulative losses, pool factors, and repayment performance.

This analysis helps compare the performance of different loan vintages and supports investment decisions.

Dynamic Loss Analysis

Dynamic Loss Analysis evaluates portfolio performance on a periodic basis, typically monthly.

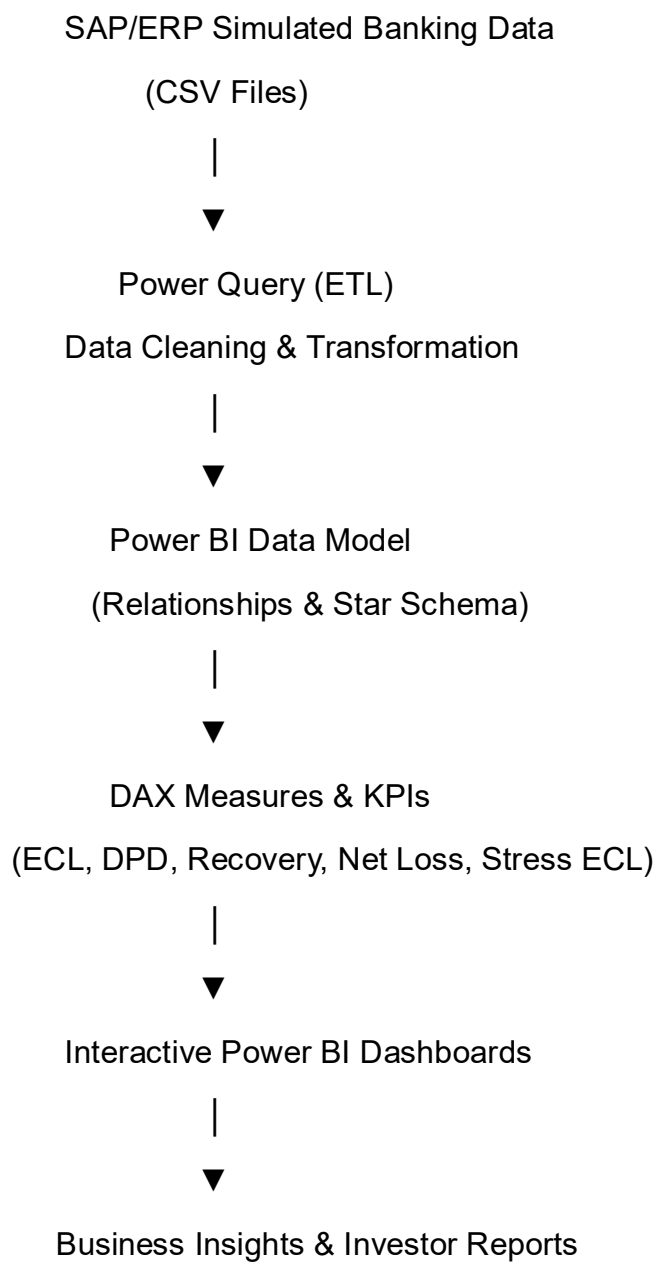
It tracks:

- Defaults

- Recoveries
- Net Losses
- Beginning of Period (BOP) Balance
- End of Period (EOP) Balance

This analysis provides insight into portfolio health and supports investor reporting.

2. Project Architecture Diagram



3. Problem Statement

Banks and financial institutions manage large portfolios of loans that are exposed to credit risk. Traditional reporting methods often provide delayed insights and limited analytical capabilities. There is a need for an integrated analytics platform that enables real-time monitoring of loan performance, Expected Credit Loss, delinquency, vintage performance, and stress testing. The objective of this project is to develop an interactive Power BI solution that transforms raw loan data into actionable business insights for effective securitisation risk management.

4. Project Objectives

The primary objectives of this project are:

- Design a securitisation risk analytics data model.
- Build an IFRS 9 Expected Credit Loss model.
- Monitor loan delinquency using DPD analysis.
- Perform static pool and vintage analysis.
- Develop dynamic loss reporting dashboards.
- Create stress testing scenarios using Power BI.
- Generate investor-friendly interactive dashboards.
- Support business decision-making through data visualization.

5. Scope

The project focuses on the analysis of an auto loan securitisation portfolio using simulated banking data. It includes data modeling, DAX calculations, dashboard development, stress testing, and investor reporting. The project does not include live integration with SAP/ERP systems or production deployment but demonstrates the architecture and analytical capabilities required for enterprise risk analytics.

6. Future Enhancements

Potential enhancements include:

- Integration with live SAP/ERP systems.
- Azure SQL Database connectivity.
- Real-time data streaming.

- Machine learning models for default prediction.
- Predictive risk scoring.
- Power BI Service deployment for enterprise sharing.
- Role-based security (Row-Level Security).
- Automated report refresh and alerting.

7. Technologies Used

Tool	Purpose
Microsoft Power BI Desktop	Dashboard development and visualization
Microsoft Excel / CSV	Source data storage
Power Query	Data cleaning and transformation
DAX (Data Analysis Expressions)	Financial calculations and measures
Data Model	Relationship management
Power BI Service (Optional)	Dashboard publishing and sharing

8. Dataset Description

Note: Replace the column counts below with the actual numbers from your dataset.

Table 1 – Auto Loan Securitisation Data

Purpose: Contains master information for each loan.

Primary Key: LoanID

Important Fields:

- LoanID
- BorrowerID
- LoanAmount
- CurrentBalance
- PD_Estimate
- LGD_Estimate
- EAD
- IFRS9_Stage

- Region
- VehicleType

Business Use: Used as the primary table for portfolio analysis and ECL calculations.

Table 2 – DPD Snapshot History

Purpose: Stores historical delinquency information.

Primary Key: LoanID + SnapshotDate

Important Fields:

- LoanID
- SnapshotDate
- DPD_Days
- CurrentBalance

Business Use: Supports delinquency analysis and roll-rate monitoring.

Table 3 – Static Pool Vintage Data

Purpose: Tracks performance of loan vintages.

Primary Key: VintageID + MonthsOnBook

Important Fields:

- VintageID
- MonthsOnBook
- PoolFactor
- CumulativeNetLossRate

Business Use: Used for vintage performance and loss curve analysis.

Table 4 – Dynamic Loss Monthly

Purpose: Monthly portfolio loss reporting.

Primary Key: ReportingDate

Important Fields:

- ReportingDate

- Defaults
- Recoveries
- NetLoss
- BOP_Balance

Business Use: Used for investor reporting and waterfall analysis.

9. Data Model

Star Schema

The project follows a Star Schema where the master loan table acts as the central dimension, connected to transaction and historical fact tables through key relationships. This structure improves query performance and simplifies analytical reporting.

Relationships:

- Auto Loan Securitisation Data → DPD Snapshot History (LoanID)
- Auto Loan Securitisation Data → Static Pool Vintage Data (VintageID/PoolID)
- Calendar Table → DPD Snapshot History (SnapshotDate)
- Calendar Table → Dynamic Loss Monthly (ReportingDate)

These relationships enable time-series analysis, drill-down capabilities, and consistent filtering across all dashboard pages.

Data Cleaning Process

The following preprocessing steps were performed using Power Query:

- Imported CSV datasets into Power BI.
- Converted date columns to Date data type.
- Formatted monetary columns as Currency/Decimal.
- Verified numeric fields for calculations.
- Checked and handled missing or null values where applicable.
- Standardized column names and data formats.
- Validated relationships before creating the data model.

DAX Documentation

Create a table like this and populate it with your actual measures:

Measure	Purpose
Total Balance	Calculates total outstanding portfolio balance
Loan Count	Counts the total number of active loans
Average PD	Calculates average Probability of Default
Average LGD	Calculates average Loss Given Default
ECL	Calculates Expected Credit Loss
Delinquency Rate	Calculates percentage of delinquent loans
Recovery Rate	Calculates recovery efficiency
Stress ECL	Calculates ECL under stress scenarios

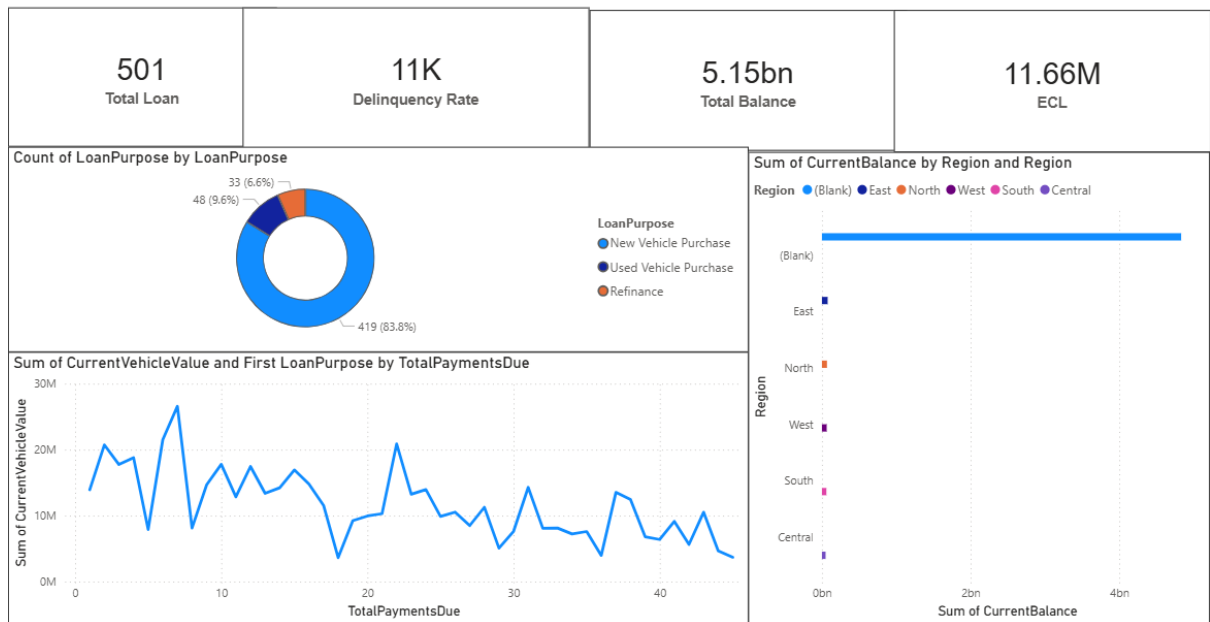
Include the actual DAX formulas from your PBIX file in a second column if required.

10. Dashboard Documentation

➤ Executive Overview

- Objective: Provide a high-level summary of portfolio performance.
- KPIs: Total Balance, Loan Count, ECL, Delinquency Rate.
- Visuals: KPI Cards, Line Chart, Donut Chart, Bar Chart.
- Business Insight: Enables management to assess portfolio size, composition, and overall risk.

Executive Overview



➤ IFRS 9 Analysis

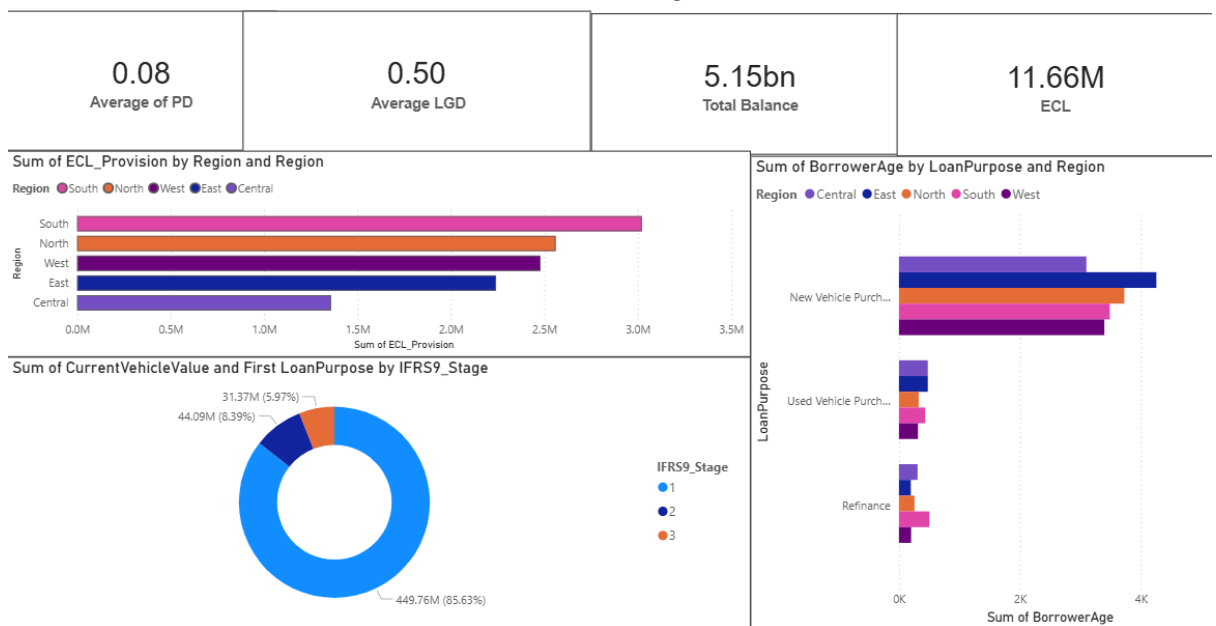
Objective: Evaluate the securitised loan portfolio using the IFRS 9 Expected Credit Loss framework.

KPIs: Average PD, Average LGD, Total Balance, Expected Credit Loss (ECL).

Visuals: KPI Cards, Clustered Bar Charts, Donut Chart.

Key Findings: The portfolio has a total balance of **5.15 Billion** with an **ECL of 11.66 Million**. The **South region** contributes the highest expected credit loss, while **Stage 1 loans account for approximately 85.63%** of the portfolio, indicating that most loans remain in good standing under the IFRS 9 impairment model.

IFRS 9 Analysis



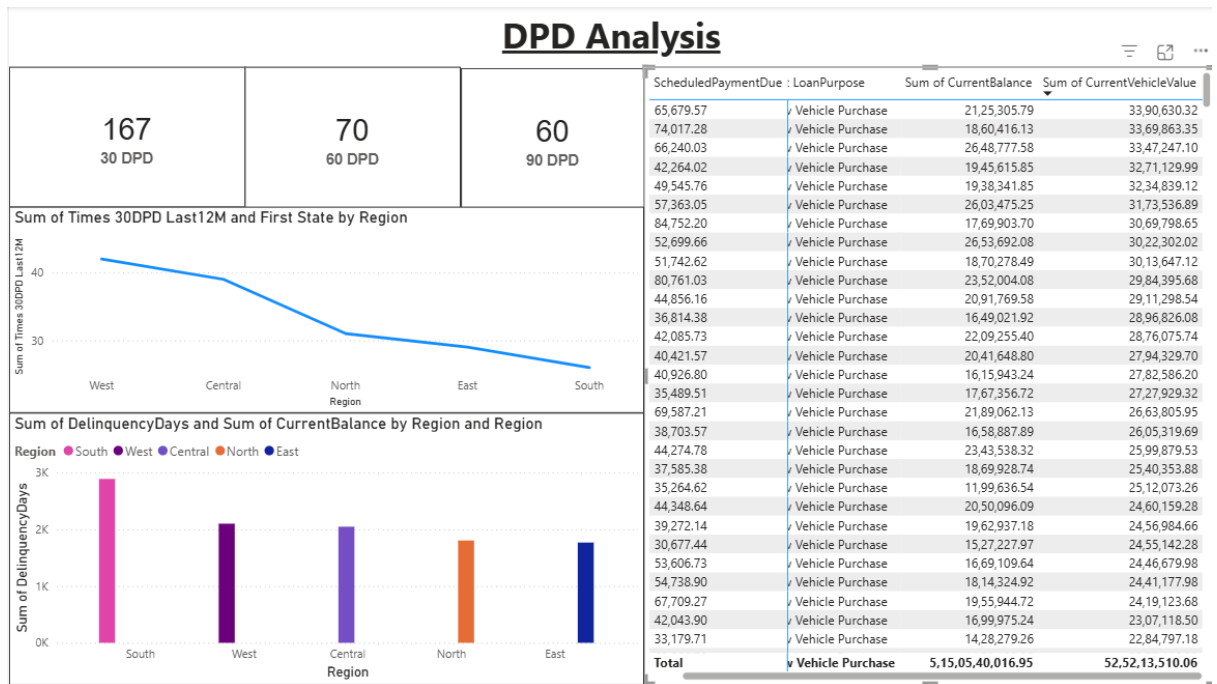
➤ DPD Dashboard

Objective: Monitor borrower delinquency and repayment behaviour using Days Past Due (DPD) analysis.

KPIs: 30 DPD (167), 60 DPD (70), 90 DPD (60).

Visuals: KPI Cards, Line Chart, Clustered Column Chart, Loan-Level Detail Table.

Key Findings: The portfolio contains **167 accounts in the 30 DPD category**, with **60 accounts reaching 90 DPD**, indicating higher default risk. The **West region** shows the highest frequency of early payment delays, while the **South region** has the highest cumulative delinquency days. The portfolio maintains **₹5.15 Billion** in outstanding balance, backed by approximately **₹5.25 Billion** in collateral value, supporting overall portfolio stability despite delinquency risks.



➤ Static Pool Dashboard

Objective: Monitor the long-term performance of loan portfolios (static pools) by analyzing loan balances, loss rates, regions, and vehicle types over time.

KPIs:

- **Total Pool Records: 6,500**
- **Average Loss Rate: 2.55K**

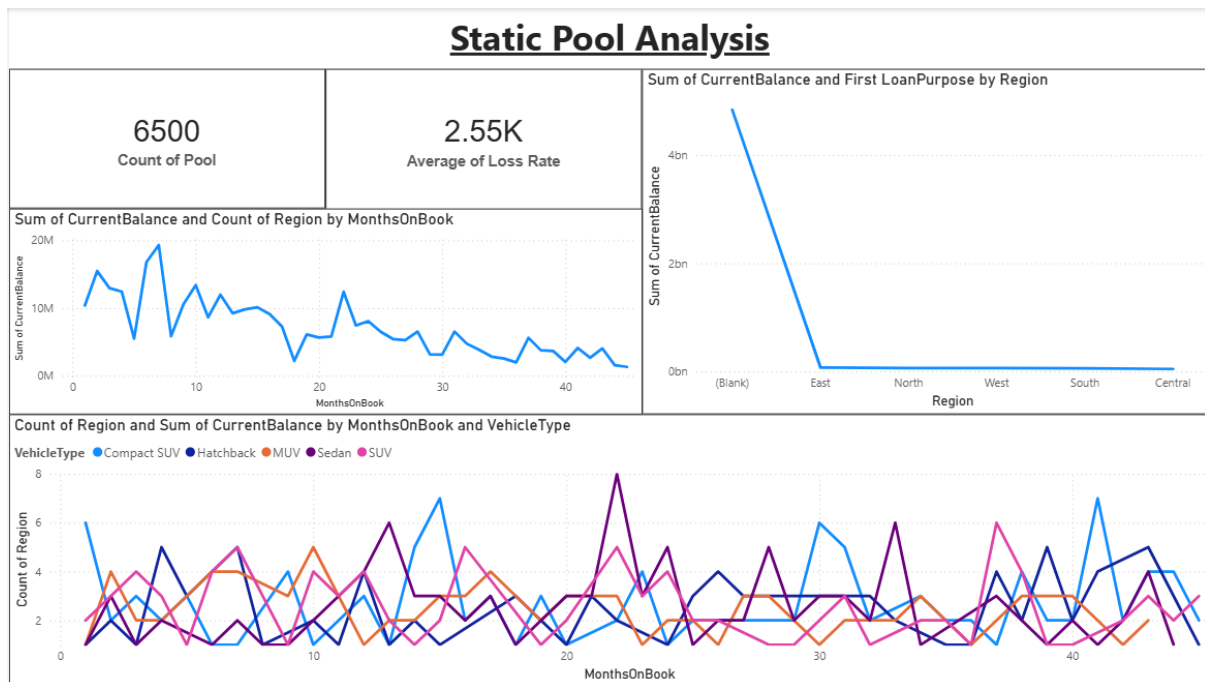
Visuals:

- KPI Cards
- Line Chart (Current Balance vs Months on Book)
- Line Chart (Current Balance by Region)
- Multi-Line Chart (Vehicle Type Performance by Months on Book)

Key Findings:

- The dashboard contains **6,500 loan records** with an **average loss rate of 2.55K**.
- The **current balance decreases as the loan age (Months on Book) increases**, showing normal loan repayments.
- One region holds the **highest outstanding loan balance**, while other regions have lower balances.
- Different **vehicle types show different repayment patterns**, with some maintaining higher balances for a longer period.

Overall, the portfolio shows **steady loan repayments and stable performance**, making it useful for monitoring long-term credit risk and loan performance.



Business Insights

Summarize the key findings from your dashboards, for example:

- Distribution of the loan portfolio across regions and loan types.
- Identification of high-risk customer segments.
- Trends in delinquency over time.
- Comparative performance of different loan vintages.
- Monthly default, recovery, and net loss behavior.
- Impact of stress scenarios on Expected Credit Loss.

Ensure these insights are based on your actual dashboard results.

Challenges Faced

Typical challenges include:

- Designing an efficient data model.
- Creating correct table relationships.
- Writing accurate DAX measures.

- Cleaning and transforming data.
- Optimizing dashboard performance.
- Implementing stress testing and interactive visuals.

11. Project Outcomes

- Developed a securitisation risk analytics dashboard.
- Implemented IFRS 9 Expected Credit Loss calculations.
- Enabled portfolio monitoring using DPD analysis.
- Created vintage and dynamic loss reporting.
- Built interactive dashboards with drill-down and filtering capabilities.
- Demonstrated stress testing for scenario-based risk assessment.

12. Conclusion

This project demonstrates how Power BI can be used to transform raw loan data into actionable financial insights. By combining data modeling, DAX calculations, and interactive dashboards, the solution supports portfolio monitoring, credit risk assessment, and investor reporting. The project also illustrates the application of IFRS 9 principles and stress testing techniques within a securitisation framework, providing a foundation for more advanced enterprise risk analytics.

13. References

Use authoritative sources such as:

- Microsoft Power BI Documentation
- Microsoft DAX Documentation
- IFRS Foundation – IFRS 9 Financial Instruments
- Basel Committee on Banking Supervision